T3000 API & Protocol

LVGL WASM Browser Build Migration · built Nov 13, 2026

LVGL WASM Browser Build Migration

Date: 2025-06-25 (updated 2026-07-10) **Project:** T3000 Webview — EEZ Studio Migration (Electron → Browser) **Branch:** WASM runtime for LVGL project editor

Overview

Migrated the EEZ Studio LVGL flow runtime WASM from Electron/Node.js target to browser (Vite/Quasar). The original `studio-wasm-libs` produces WASM modules targeting Node.js APIs (`__dirname` , `NODEJS_CATCH_EXIT`). These were rebuilt with Emscripten targeting `ENVIRONMENT=web` and deployed as static assets served by the Rust backend.

Two WASM runtimes are now served:

Runtime	Served at	Used by
LVGL per-version	`/eez-studio-wasm/wasm/lvgl/{version}/`	LVGL project page rendering (lvgl-versions.ts)
Dashboard (eez_runtime)	`/eez-studio-wasm/wasm/`	Flow runtime for dashboards + GUI lite (wasm-worker.ts)

Architecture

TEXT

```
┌──────────────────────────────────────────────┐
│ Browser (Vite/Quasar)                        │
│                                              │
│  lvgl-versions.ts                            │
│    │ detect window !== undefined             │
│    │ construct URL:                          │
│    │   /eez-studio-wasm/wasm/lvgl/{v}/...    │
│    ▼                                         │
│  <script src=...> → WebAssembly.instantiate()  │
│    │                                         │
│  LVGL Runtime (Module._lvglInit, _lvglTick...)  │
│                                              │
│  wasm-worker.ts (Web Worker)                 │
│    │ eez_runtime.js, eez_runtime.wasm        │
│    │   /eez-studio-wasm/wasm/                │
│    ▼                                         │
│  Flow Runtime (dashboard + GUI lite)         │
└──────────────────────────────────────────────┘
        │
        │ HTTP GET (dev: Vite proxy → Rust; prod: Rust serve)
        ▼
┌──────────────────────────────────────────────┐
│ Rust Backend (port 9103)                     │
│                                              │
│  routes_wasm() — ServeDir at:                │
│    T3Web/t3-eez/resources/eez-studio-wasm/   │
│                                              │
│  Contents (copied by build.rs):              │
│   wasm/                                      │
│   ├── eez_runtime.{js,wasm}                  │
│   ├── eez_gui_lite_runtime.{js,wasm}         │
│   ├── lz4.{js,wasm}                          │
│   └── lvgl/                                  │
│       ├── 8.4.0/lvgl_runtime_v8.4.0.{js,wasm}  │
│       ├── 9.2.2/lvgl_runtime_v9.2.2.{js,wasm}  │
│       ├── 9.3.0/lvgl_runtime_v9.3.0.{js,wasm}  │
│       ├── 9.4.0/lvgl_runtime_v9.4.0.{js,wasm}  │
```

```
│       └── 9.5.0/lvgl_runtime_v9.5.0.{js,wasm}   │
└──────────────────────────────────────────────────┘
```

Source Repository

`https://github.com/eez-open/studio-wasm-libs`

Local clone: `C:\QN\temcocontrols\studio-wasm-libs`

Submodules

Submodule	Path	Version
`eez-framework`	`eez-framework/`	master
`lvgl`	`lvgl-runtime/v8.4.0/lvgl`	v8.4.0
`lvgl`	`lvgl-runtime/v9.2.2/lvgl`	v9.2.2
`lvgl`	`lvgl-runtime/v9.3.0/lvgl`	v9.3.0
`lvgl`	`lvgl-runtime/v9.4.0/lvgl`	v9.4.0
`lvgl`	`lvgl-runtime/v9.5.0/lvgl`	v9.5.0

Patches Applied

1. `lvgl-runtime/common/pre.js` — Browser file resolution

DIFF

```
- return __dirname + "/" + path;
+ return new URL(path, import.meta.url).href;
```

`__dirname` is a Node.js global. Browsers use `import.meta.url`.

2. All 5 `lv_conf.h` files — Disable FreeType

DIFF

```
- #define LV_USE_FREETYPE 1
+ #define LV_USE_FREETYPE 0
```

FreeType requires system font loading (`/home/mvladic/freetype-2.14.1`) not available in browser WASM.

Files: `lvgl-runtime/{v8.4.0,v9.2.2,v9.3.0,v9.4.0,v9.5.0}/lv_conf.h`

3. All 5 `CMakeLists.txt` — Remove hardcoded FreeType paths

Removed:

- `-L/home/mvladic/freetype-2.14.1/build -lfreetype` from linker flags
- `include_directories(/home/mvladic/freetype-2.14.1/include)` line

Files: `lvgl-runtime/{v8.4.0,v9.2.2,v9.3.0,v9.4.0,v9.5.0}/CMakeLists.txt`

4. `lvgl-runtime/common/src/studio_api.cpp` — Guard FreeType code

Wrapped `lvglCreateFreeTypeFont()` with `#if LV_USE_FREETYPE` / `#endif` preprocessor guards to prevent compilation errors from `lv_ft_info_t` usage in v8.x code path.

5. `lvgl-runtime/v9.5.0/exported-functions.txt` — Remove FreeType exports

Removed 17 `_lv_freetype_*` symbols from the Emscripten export list. These functions don't exist when FreeType is disabled, causing `wasm-ld` linker errors.

6. `lvgl-runtime/common/src/flow.cpp` — Remove `setNotStopped()`

Removed `eez::flow::setNotStopped()` call. Upstream `eez-framework` removed this function; `g_isStopped` defaults to `true` and is set to `false` by `start()` (called during `_init()`), making the call redundant.

Build Environment

Tool	Version	Path
Emscripten SDK	6.0.1	`C:\QN\temcocontrols\emsdk`
CMake	4.3.3	`C:\Program Files\CMake\bin`
Ninja	1.13.2	`C:\QN\temcocontrols\ninja`

Build Command

```cmd
call C:\QN\temcocontrols\emsdk\emsdk_env.bat
set PATH=C:\Program Files\CMake\bin;C:\QN\temcocontrols\ninja;%PATH%
cd lvgl-runtime\<version>
mkdir build && cd build
emcmake cmake ..
emmake ninja -j4
```

Build Results

Version	JS	WASM	Files
v8.4.0	608 KB	1.96 MB	238 source files
v9.2.2	763 KB	1.97 MB	476 source files
v9.3.0	815 KB	2.10 MB	584 source files
v9.4.0	856 KB	2.14 MB	691 source files
v9.5.0	889 KB	2.18 MB	694 source files

Build & Deploy Pipeline

TEXT

```
1. Edit C++ source in studio-wasm-libs/
       │
2. Build all: cd studio-wasm-libs && .\build-all.bat
       │  Output → studio-wasm-libs/release/wasm/
       │    eez_runtime.{js,wasm}
       │    eez_gui_lite_runtime.{js,wasm}
       │    lz4.{js,wasm}
       │    lvgl_runtime_v{8.4.0,9.2.2,9.3.0,9.4.0,9.5.0}.{js,wasm}
       │
3. cargo build (or cargo build --release)
       │  build.rs auto-copies from studio-wasm-libs/release/wasm/
       │  → target/<profile>/T3Web/t3-eez/resources/eez-studio-wasm/wasm/
       │
4. Rust server serves at runtime
       │  /eez-studio-wasm/wasm/... → T3Web/t3-eez/resources/eez-studio-wasm/
```

> **Never manually copy files into `api/target/`**. The pipeline is: edit in `studio-wasm-libs` → `build-all.bat` → `cargo build` (auto-copy) → serve.

Frontend Code Changes

`lvgl-versions.ts` — LVGL runtime URL

DIFF

```
- const jsUrl = `/t3-eez-studio/wasm/lvgl/${version}/${fileName}`;
+ const jsUrl = `/eez-studio-wasm/wasm/lvgl/${version}/${fileName}`;
```

Also the v8.4.0 hardcoded fallback path:

DIFF

```
- wasmFlowRuntime: "project-editor/flow/runtime/wasm/lvgl_runtime_v8.4.0.js",
+ wasmFlowRuntime: "/eez-studio-wasm/wasm/lvgl/8.4.0/lvgl_runtime_v8.4.0.js",
```

`wasm-worker.ts` — Dashboard flow runtime (Web Worker)

```typescript
const wasmPath = "/eez-studio-wasm/wasm/eez_runtime.wasm";
(g as any).__dirname = "/eez-studio-wasm/wasm";
s.src = `/eez-studio-wasm/wasm/eez_runtime.js?v=${EEZ_RUNTIME_CACHE_BUSTER}`;
```

Vite Configuration

Dev proxy (`quasar.config.js`)

In development, Vite (port 3003) proxies WASM requests to the Rust backend (port 9103):

```javascript
viteConf.server.proxy = {
  "/eez-studio-wasm": {
    target: "http://localhost:9103",
    changeOrigin: true,
  },
  "/eez-studio-assets": {
    target: "http://localhost:9103",
    changeOrigin: true,
  },
};
```

WASM MIME type

```javascript
viteConf.server.headers = { "*.wasm": { "Content-Type": "application/wasm" } };
```

Ensures Vite dev server serves `.wasm` files with correct MIME type for `WebAssembly.instantiateStreaming()`.

Runtime Loading Flow

LVGL Runtime (lvgl-versions.ts)

1. User opens an LVGL project page in the EEZ Studio editor

2. `page-runtime.ts` creates a runtime instance
3. `lvgl-versions.ts:getLvglWasmFlowRuntimeConstructor()` is called
4. Detects `typeof window !== "undefined"` (browser environment)
5. Extracts version from `wasmFlowRuntime` property (e.g., `"9.5.0"`)
6. Constructs URL: `/eez-studio-wasm/wasm/lvgl/<version>/lvgl_runtime_v<version>.js`
7. Loads Emscripten JS glue via `<script>` tag → sets `globalThis.LVGLWasmRuntime`
8. WASM module exports LVGL functions (`_lvglInit`, `_lvglTick`, etc.)
9. Fallback: Proxy-based no-op if WASM fails to load (prevents crashes)

Dashboard Flow Runtime (wasm-worker.ts)

1. Web Worker loads `eez_runtime.js` via `importScripts()`
2. `__dirname` set to `/eez-studio-wasm/wasm` for Emscripten's `locateFile`
3. `WebAssembly.instantiateStreaming(fetch(wasmPath))` loads `eez_runtime.wasm`
4. Worker communicates with main thread via `postMessage` (action-based protocol)
5. Flow runtime executes dashboard logic in the worker, sends rendering commands to main thread

---

Path Rename: `t3-eez-studio` → `eez-studio-wasm`

All WASM-related URLs were renamed to match the Rust-side naming:

File	Change
`lvgl-versions.ts` (line 199)	Hardcoded v8.4.0 path
`lvgl-versions.ts` (line 613)	Dynamic URL construction
`wasm-worker.ts` (lines 51, 82, 85)	`wasmPath`, `__dirname`, script `src`
`quasar.config.js` (line 231)	Vite proxy entry
`api/build.rs` (3 calls)	`copy_selected_files` dest_rel → `"eez-studio-wasm"`
`api/src/server.rs` (lines 131, 254)	`ServeDir` path + `.nest("/eez-studio-wasm", ...)`

Known Limitations

1. **FreeType disabled** — Custom TTF/OTF fonts not supported. Only built-in LVGL bitmap fonts (Montserrat).
2. **No file system** — Emscripten FS not initialized in browser. Image loading via memory buffers only.

Future Rebuilds

When `studio-wasm-libs` upstream updates:

1. `git pull` + `git submodule update --init --recursive`
2. Re-apply patches 1-6 above
3. Run `build-all.bat` (builds all 5 LVGL versions + eez_runtime + eez_gui_lite_runtime + lz4)
4. `cargo build` — `build.rs` auto-copies WASM files to target
5. Restart Rust backend to serve updated files